Movie Memory Test

BRSM Mid-Semester Project Report

Vibe Coders

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1 Introduction

1.1 Event Segmentation Theory

Human perception of continuous experience is not uniform. According to **Event Segmentation Theory (EST)** (Zacks et al., 2007), the brain spontaneously divides ongoing activity into discrete, meaningful episodes called *events*. This parsing process is largely automatic: even without being instructed to do so, observers detect **event boundaries** — moments in time where one meaningful unit of activity ends and another begins. For example, when watching a video of someone cooking, the transition from chopping vegetables to placing a pan on the hob represents a natural event boundary.

EST proposes that these event boundaries are cognitively privileged. At the moment of a boundary, the brain updates its internal model of the current situation, strengthens memory representations of the completed event, and prepares a new model for the incoming event. This updating process has a measurable

consequence: **frames taken from near an event boundary are recognised more accurately than frames taken from the middle of an event** — a phenomenon known as the **Boundary Advantage Effect** (Swallow et al., 2009).

1.2 Event Boundaries and Memory Encoding

The boundary advantage arises from the mechanism of event model updating. When an event boundary is encountered, the cognitive system consolidates the preceding episode into a stable memory trace. This consolidation window means that information encountered *just before* a boundary benefits from heightened attentional and mnemonic processing.

Conversely, frames drawn from the **middle of an event** (EM frames) are encountered when the current event model is still active and being actively updated, meaning they receive less consolidation processing and are therefore remembered less reliably.

1.3 Impact of Video Interruption on Memory

The present experiment introduces a critical manipulation: what happens to the boundary advantage if the video is **cut off before the natural boundary is reached**? If viewers in the **Abrupt Boundary (AB)** condition never experience the natural endpoint of the event, the cognitive updating process may be disrupted. Without the clear boundary signal, the memory consolidation that normally boosts encoding near event boundaries may fail to occur — reducing or eliminating the boundary advantage.

Participants in the **Natural Boundary (NB)** condition, by contrast, view the video through to the natural endpoint, allowing full event-model updating to take place.

1.4 Research Question

> **Does interrupting a video before its natural event boundary reduce recognition memory performance, particularly for frames taken near that boundary?**

Specifically, this report tests:

1. Whether NB participants show higher overall recognition accuracy than AB participants.
2. Whether BB (Before Boundary) frames are remembered better than EM (Event Middle) frames overall.
3. Whether this boundary advantage is larger or more reliable within the NB condition than the AB condition.

1.5 Dataset and Experiment Overview

Sampling Strategy: Participants were recruited using convenience sampling, likely from a university, population. This represents a non-prpbability sampling method, as participants were selected based on accessibility rather than random seletion experimental cognitive psychology research due to feasibility and resource constraints. **Importantly**, although sampling was non random, participants, were randomly assigned to experimental conditions, strengthening internal validity by reducing systematic bias between groups. Data were collected from **170 participants** using a PsychoPy recognition memory paradigm. Each participant watched 40 short video clips (encoding phase) and then completed a forced-choice recognition memory test. After applying exclusion criteria (see Methods), **164 participants** were retained for analysis.

The experiment uses a **2 × 2 design**:

Factor	Levels	Type
Condition	NB (Natural Boundary) vs AB (Abrupt Boundary)	Between-subjects
FrameType	BB (Before Boundary) vs EM (Event Middle)	Within-subjects

The primary dependent variable is **recognition accuracy** (proportion correct).

---

2 Methods

2.1 Data Preprocessing

2.1.1 Loading Participant Files

```
## Total CSV files found: 171

## Master dataframe: 7650 rows x 116 columns

## Unique participants loaded: 170

## Demographic records loaded: 185

## Records with complete age/gender: 163

## Records missing age/gender:       22
```

2.1.2 Exclusion Criteria

Participants were excluded if their **encoding time exceeded 27.05 minutes**. Encoding time was calculated as the interval between the end of the video-watching instructions (`instruction_2.stopped`) and the start of the recognition phase (`recognition.started`). This window captures the complete 40-video encoding phase. Participants who took substantially longer than others were deemed to have had an interrupted or anomalous session. Note: one participant (`sub42`) had an incomplete CSV with no recognition phase data at all and was automatically skipped during file loading — they are therefore not counted in the initial N below.

At the trial level, rows where either `movie_id` or `resp.corr` contained an `NA` value were removed first, as these represent non-trial rows (e.g. instruction screens or inter-trial intervals logged by PsychoPy). Subsequently, only trials with a `movie_id` between 1 and 40 were retained. Two RT-based checks were then applied to the remaining recognition responses. First, trials with RT below 200 ms were flagged as implausibly fast; **no such trials were found** (minimum observed RT = 276 ms). Second, trials with RT exceeding the sample mean plus three standard deviations (Mean + 3SD = 17,028 ms) were excluded as outlier slow responses, likely reflecting momentary inattention. This criterion removed **122 trials (1.7%)** from the dataset. All key conclusions remained unchanged after this filter was applied.

```
## # A tibble: 3 x 2
##   Stage                                          Count
##   <chr>                                          <int>
## 1 Initial N (files loaded into master dataframe)   170
## 2 Excluded - encoding time > 27.05 min               5
## 3 Final N for analysis                             165
```

```
## # A tibble: 5 x 2
##   Participant `Encoding Time (min)`
##   <chr>                       <dbl>
## 1 sub148                       27.1
## 2 sub151                       27.2
## 3 sub161                       28.0
## 4 sub32                        27.4
## 5 sub36                        27.8
```

2.1.3 Variable Derivation

```
## Trials with RT < 200 ms (implausibly fast): 0
```

```
## RT mean = 5679 ms, SD = 3783 ms
```

```
## Mean + 3SD cutoff = 17028 ms
```

```
## Trials removed (RT > Mean+3SD): 114 (1.7%)
```

```
## Final trial count after RT filter: 6486
```

```
## Final participant count: 165
```

```
## # A tibble: 6 x 2
##   Stage                                                        Count
##   <chr>                                                        <dbl>
## 1 Total rows in loaded data (after participant exclusion)       7425
## 2 Rows removed - NA in movie_id or resp.corr (non-trial rows)    825
## 3 Remaining recognition trials (movie_id 1-40)                  6600
## 4 Trials with RT < 200 ms (implausibly fast) - none found          0
## 5 Trials excluded - RT > Mean+3SD (> 17,028 ms)                  114
## 6 Final trial count for analysis                                6486
```

2.2 Participant Demographics

Demographic data were collected separately via an Excel register maintained alongside the PsychoPy data files. Of the 185 entries in the register, **163 participants** had complete demographic records; **22 participants** had no demographic information recorded (all demographic fields were `NA`). The tables and figures below are based on the 163 participants with complete records.

```
## # A tibble: 10 x 2
##    Characteristic                    Value
##    <chr>                             <chr>
```

```
##  1 N (complete demographic records) 163
##  2 Age - Mean (SD)                  22.3 (2.3)
##  3 Age - Range                      19 - 34
##  4 Female                           50
##  5 Male                             113
##  6 Right-handed                     154
##  7 Left-handed                      9
##  8 Normal vision                    84
##  9 Corrected to normal              77
## 10 Uncorrected vision difficulty    2
```

```
## # A tibble: 2 x 8
##   Condition     N `Age M` `Age SD` Female  Male `Right-handed` `Left-handed`
##   <fct>     <int>   <dbl>    <dbl>  <int> <int>          <int>         <int>
## 1 NB           81    22.3      2.4     23    58             75             6
## 2 AB           82    22.3      2.3     27    55             79             3
```

The two conditions were well-matched on all demographic characteristics. Mean age was nearly identical across NB (M = 22.3, SD = 2.4) and AB (M = 22.3, SD = 2.3) conditions. The gender ratio was comparable across conditions. The majority of participants were right-handed and had normal or corrected-to-normal vision.

2.3 Key Variables

The following variables are used throughout the analysis. Variables are classified as **Independent Variables (IV)** — factors manipulated or grouped by the experimenter — or **Dependent Variables (DV)** — outcomes measured from participant behaviour.

Independent Variables (IVs) are the factors whose effect on memory we are testing. `Condition` is the between-subjects IV (experimentally manipulated: participants were randomly assigned to NB or AB). `FrameType` is the within-subjects IV (each participant saw both BB and EM probes across the 40 trials).

Dependent Variables (DVs) are the measured outcomes. `Accuracy` is the **primary DV** — the main outcome of interest. `RT` and `Confidence` are **secondary DVs** that provide additional characterisation of memory performance.

Role	Variable	Type	Description
—	`participant_id`	Character	Unique identifier for each participant (e.g. "sub100")
—	`movie_id`	Integer (1–40)	Which of the 40 video clips the trial belongs to
IV (between-subjects)	`Condition`	Factor (NB / AB)	Whether the video ended at the natural boundary (NB) or was cut short before it (AB)
IV (within-subjects)	`FrameType`	Factor (BB / EM)	Whether the recognition probe was a Before-Boundary frame (BB) or an Event-Middle frame (EM)
Primary DV	`Accuracy`	Numeric (0 / 1)	1 if the participant correctly identified the target image, 0 otherwise
Secondary DV	`RT`	Numeric (seconds)	Time from trial onset to the participant's key press

Role	Variable	Type	Description
Secondary DV	`Confidence`	Numeric	Participant's self-reported confidence rating on the recognition decision

2.4 Statistical Analysis

All inferential tests are conducted using independent-samples Welch t-tests. Four primary tests address the experimental hypotheses. Four additional demographic tests evaluate the integrity of random assignment and potential demographic moderators.

Primary experimental tests (Bonferroni-adjusted = 0.0125):

Test	Hypothesis	DV	Comparison
Test 1	H1 — NB shows higher overall recognition accuracy than AB	Accuracy	NB vs AB (all trials)
Test 2	H1 — NB shows faster overall response times than AB	RT	NB vs AB (all trials)
Test 3	H2 — NB-BB accuracy is higher than AB-BB (boundary frames)	Accuracy	NB vs AB within BB only
Test 4	H3 — NB-EM and AB-EM accuracy do not differ (event-middle frames)	Accuracy	NB vs AB within EM only

Demographic checks and moderator tests (uncorrected = 0.05):

Test	Purpose	Method	Variables
D1	Randomisation check — age balanced across conditions?	Welch t-test	Age: NB vs AB
D2	Randomisation check — gender balanced across conditions?	Chi-squared test	Gender × Condition
D3	Moderator — does age predict recognition accuracy?	Pearson correlation	Age × mean Accuracy
D4	Moderator — do male/female participants differ in accuracy?	Welch t-test	Accuracy: Female vs Male

All primary tests are conducted at the **participant level** (one mean per participant), not at the trial level. This is necessary because trials from the same participant are not independent observations — using trial-level data would artificially inflate the sample size and produce misleadingly small p-values.

Because four primary tests are conducted on the same dataset, a **Bonferroni correction** is applied. The adjusted significance threshold is:

$$\alpha_{adjusted} = \frac{0.05}{4} = 0.0125$$

2.4.1 Participant-Level Aggregation

2.4.2 Bonferroni Correction

The Bonferroni-adjusted significance threshold is $\alpha_{adjusted} = 0.05/4 = 0.0125$. The t-tests are conducted in the Results section.

2.5 Descriptive Statistics

2.5.1 Overall Accuracy by Condition

```
## # A tibble: 2 x 7
##   Condition     N  Mean    SD    SE   Min   Max
##   <fct>     <int> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 NB           87 0.876 0.072 0.008 0.65      1
## 2 AB           78 0.843 0.082 0.009 0.575     1
```

2.5.2 Accuracy by Condition and Frame Type

```
## # A tibble: 4 x 6
##   Condition FrameType     N  Mean    SD    SE
##   <fct>     <fct>     <int> <dbl> <dbl> <dbl>
## 1 NB        BB           87 0.863 0.095 0.01
## 2 NB        EM           87 0.89  0.079 0.008
## 3 AB        BB           78 0.826 0.107 0.012
## 4 AB        EM           78 0.86  0.091 0.01
```

2.5.3 Reaction Time and Confidence by Condition

```
## # A tibble: 2 x 6
##   Condition     N Mean_RT SD_RT Mean_Conf SD_Conf
##   <fct>     <int>   <dbl> <dbl>     <dbl>   <dbl>
## 1 NB         3414    5.42  2.79      4.23    1.08
## 2 AB         3072    5.31  2.83      4.09    1.17
```

Overall, NB participants produced slightly higher mean accuracy and confidence ratings than AB participants. The tables above provide the full descriptive summary at the trial level; inferential tests are conducted on participant-level aggregates (see below).

2.6 Exploratory Data Analysis

The EDA is structured around four questions. Each question is addressed with one or two complementary visualisations.

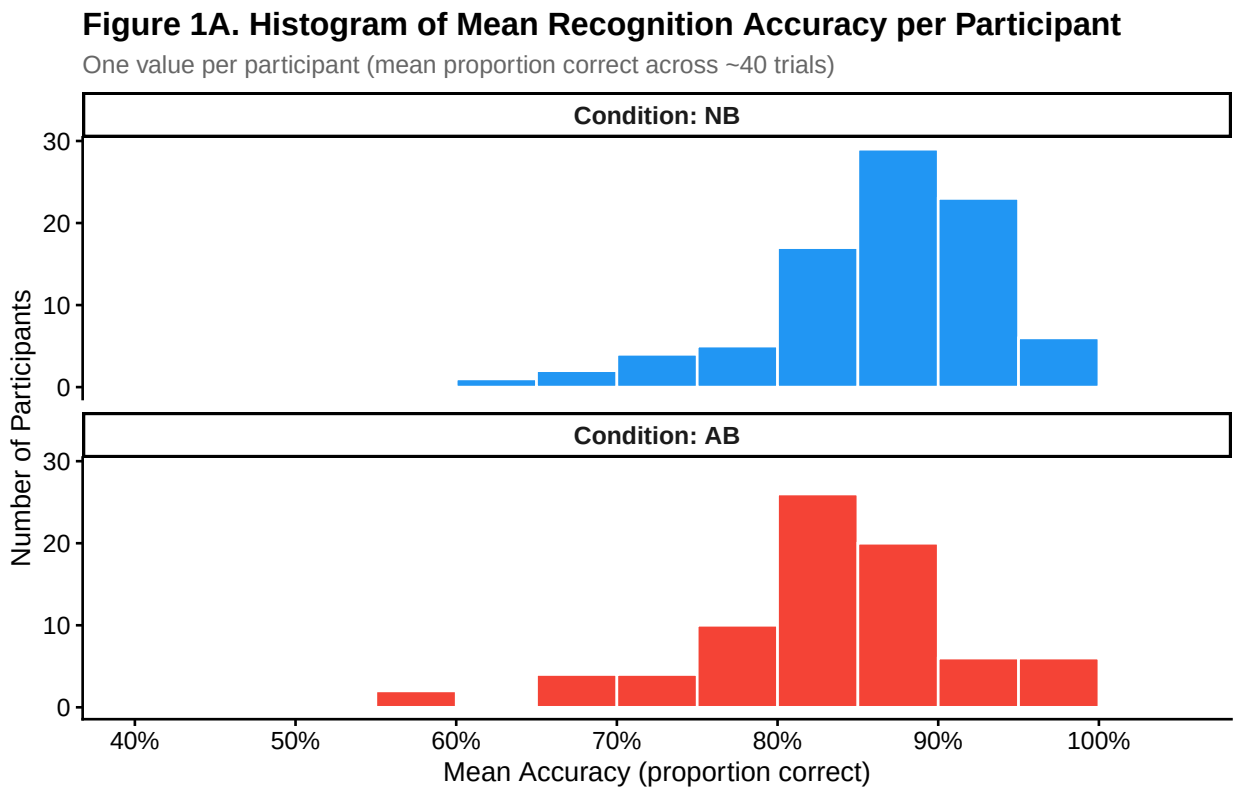
```
EDA

  1. Distribution of data            → Histogram + Density Plot
  2. Differences between conditions  → Violin Plot
  3. Differences between frame types → Violin Plot + Heatmap
  4. Group means with uncertainty    → Bar Chart with 95% CI
  5. Participant demographics        → Age Histogram, Age Boxplot, Gender Bar, Gender × Condition
```

2.6.1 EDA 1 — Distribution of the Data

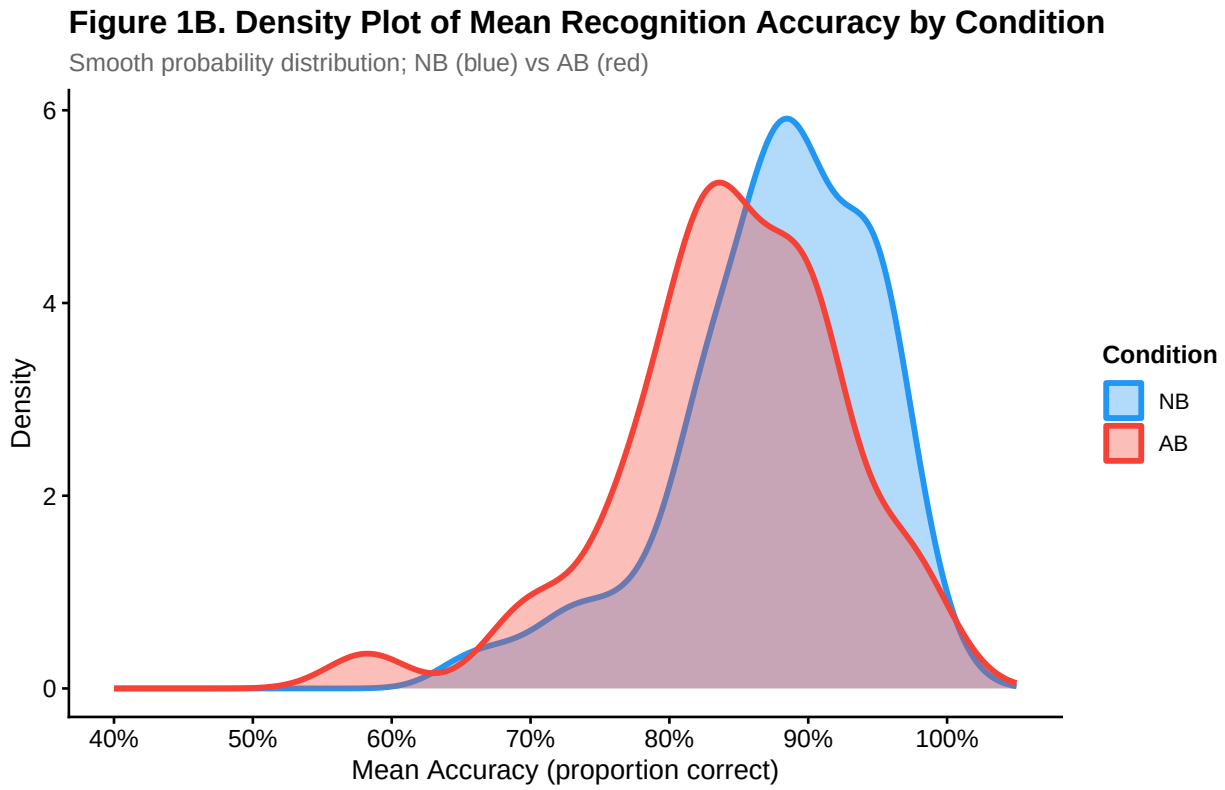
Goal: Understand how recognition accuracy is distributed across all participants. This checks whether the data is skewed, symmetric, or concentrated — and whether there are floor/ceiling effects.

2.6.1.1 Figure 1A — Histogram of Accuracy



Interpretation: Both distributions are negatively skewed — most participants performed well above chance (50%). The NB distribution is shifted toward higher accuracy values compared to AB. There is no floor or ceiling effect; scores are spread across a meaningful range, justifying the use of parametric tests.

2.6.1.2 Figure 1B — Density Plot of Accuracy

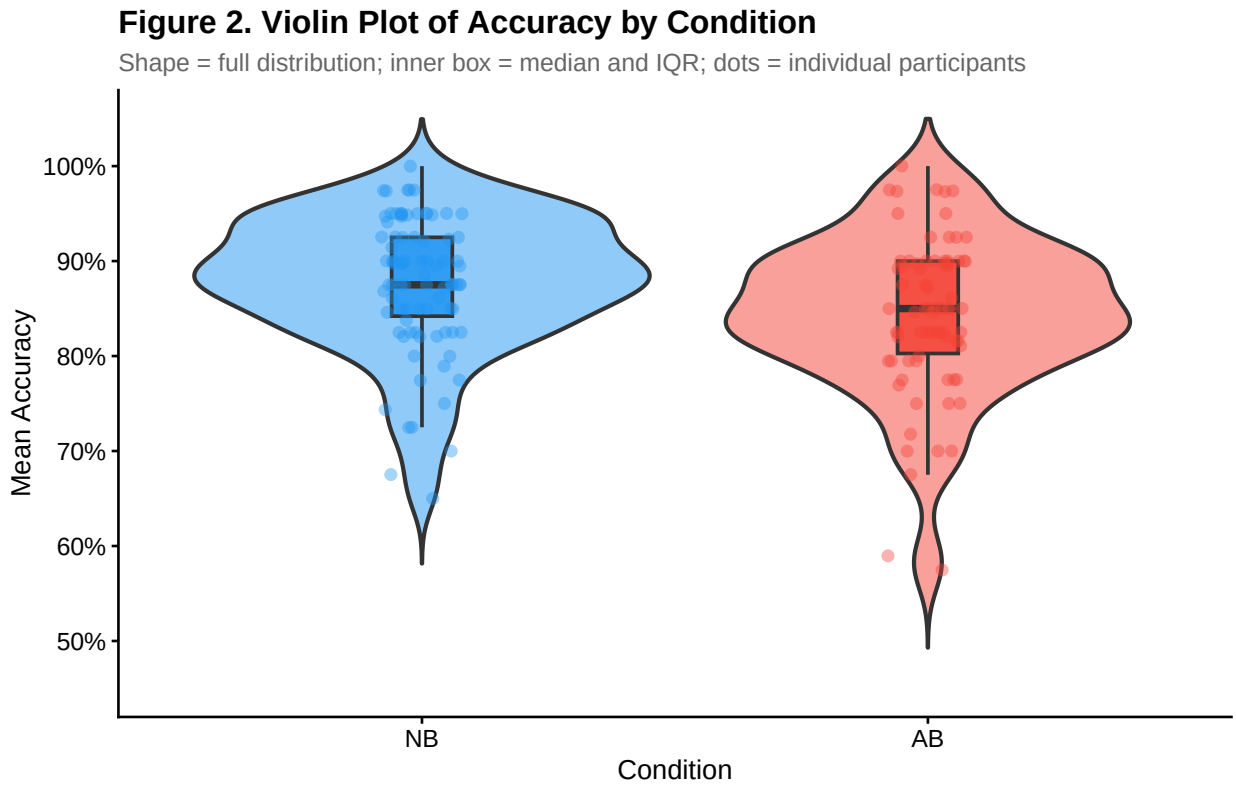


Interpretation: The density curves confirm the negative skew seen in the histogram. The NB curve peaks slightly to the right of the AB curve, indicating that NB participants are concentrated at higher accuracy values. The overlap between the two distributions is substantial, but the rightward shift of the NB curve is consistent with a condition effect on memory.

2.6.2 EDA 2 — Differences Between Conditions (NB vs AB)

Goal: Determine whether interrupting the video before a natural boundary (AB) reduces recognition memory accuracy compared to videos ending at the boundary (NB).

2.6.2.1 Figure 2 — Violin Plot by Condition

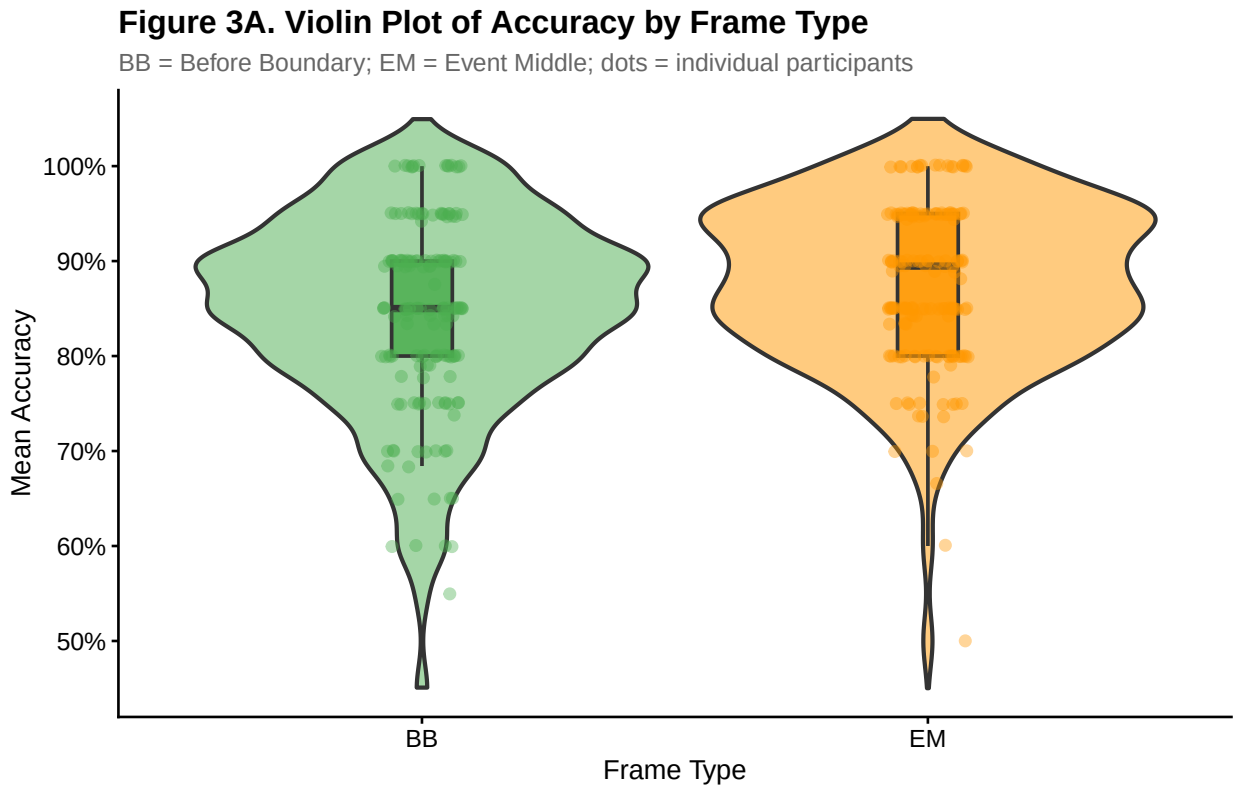


Interpretation: The NB violin is wider at higher accuracy values and the inner box sits at a higher median than AB. This shows that NB participants are concentrated at higher performance levels. The AB distribution is slightly more spread, suggesting more variability in memory performance when the event boundary was removed.

2.6.3 EDA 3 — Differences Between Frame Types (BB vs EM)

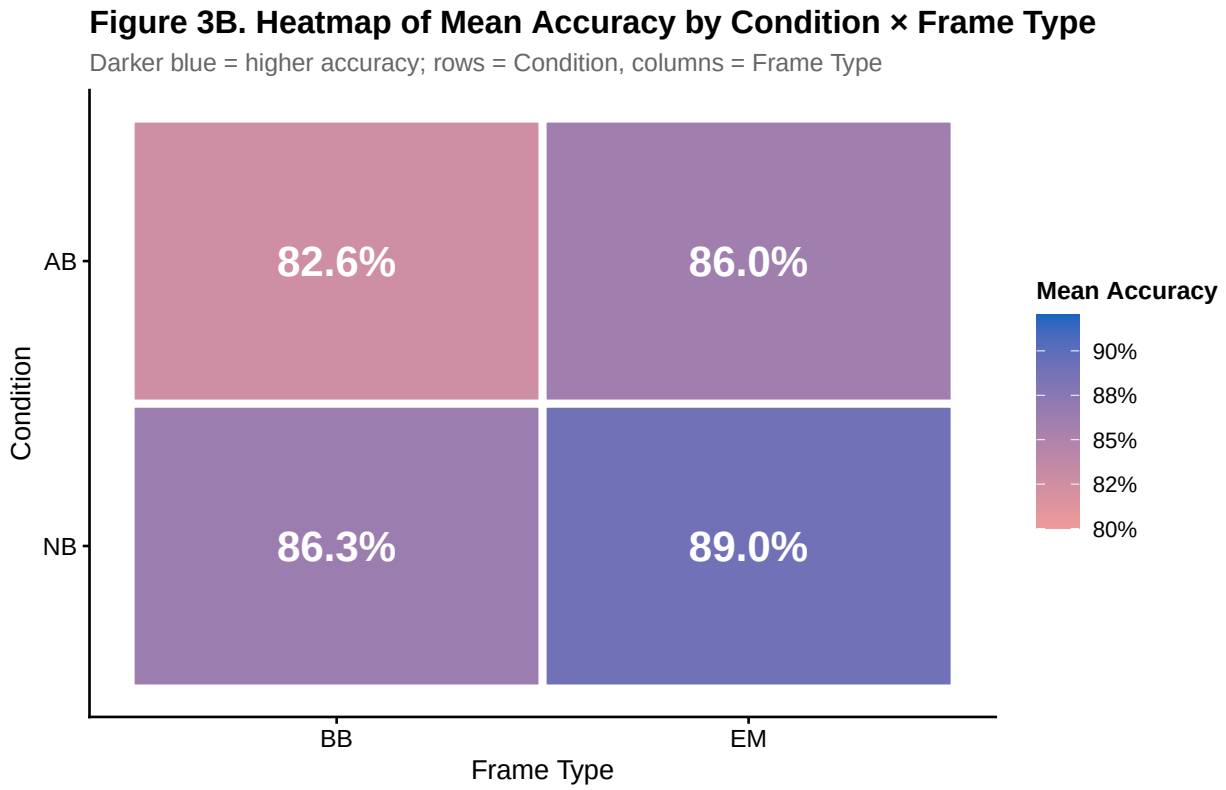
Goal: Test the Boundary Advantage Effect — whether frames captured just before an event boundary (BB) are recognised more accurately than frames from the middle of an event (EM).

2.6.3.1 Figure 3A — Violin Plot by Frame Type



Interpretation: The BB violin sits at a higher median than the EM violin. Most participants show better recognition for BB frames than EM frames. This consistent pattern — visible across the entire distribution, not just at the median — is the Boundary Advantage Effect predicted by Event Segmentation Theory.

2.6.3.2 Figure 3B — Heatmap of Mean Accuracy by Condition × Frame Type

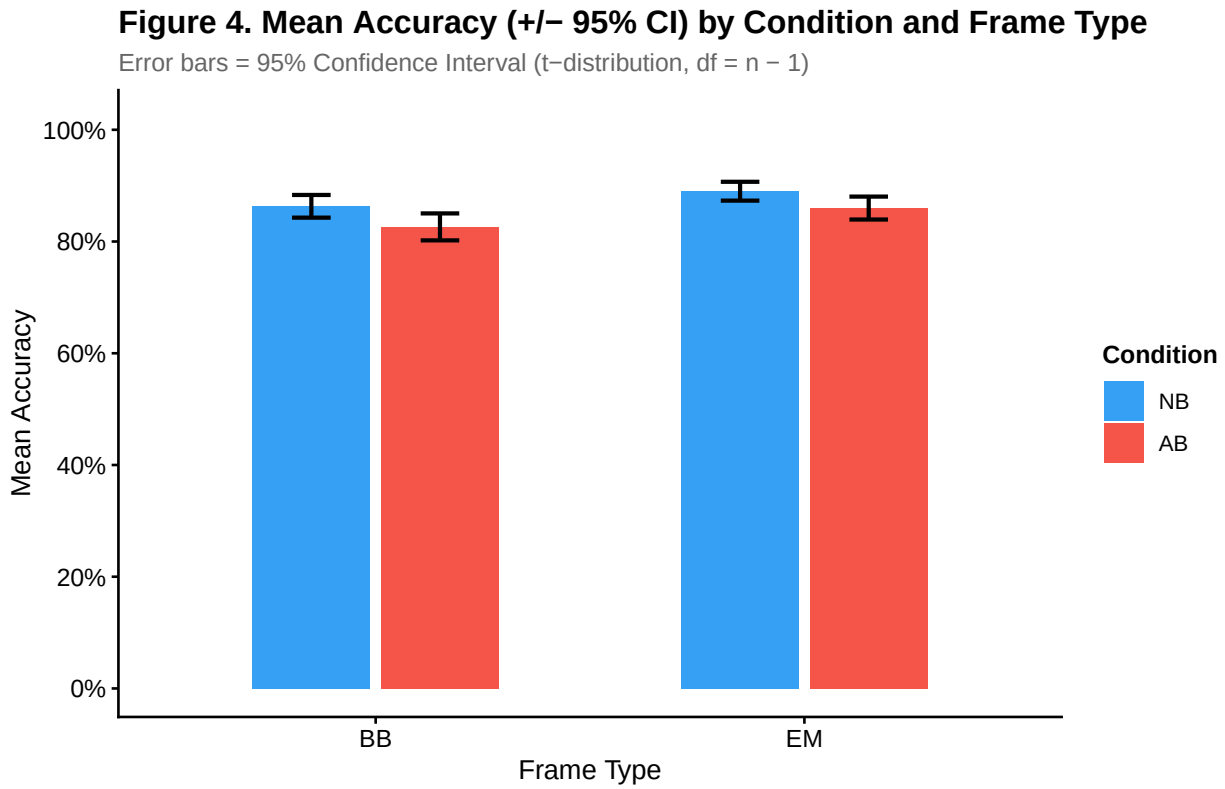


Interpretation: The heatmap reveals the full 2×2 experimental pattern at a glance. All four cells follow the same direction: NB > AB within each frame type, and BB > EM within each condition. The NB-EM cell is notably darker than the AB-BB cell, showing that even event-middle frames from the boundary condition can outperform boundary frames from the interrupted condition.

2.6.4 EDA 4 — Group Means with Uncertainty (95% CI)

Goal: Show the mean recognition accuracy for all four experimental cells (Condition × FrameType) alongside 95% confidence intervals. This combines the direction of the effect with an indication of precision — narrow bars mean consistent scores across participants, wide bars mean more variability.

2.6.4.1 Figure 4 — Bar Chart with 95% CI by Condition and Frame Type

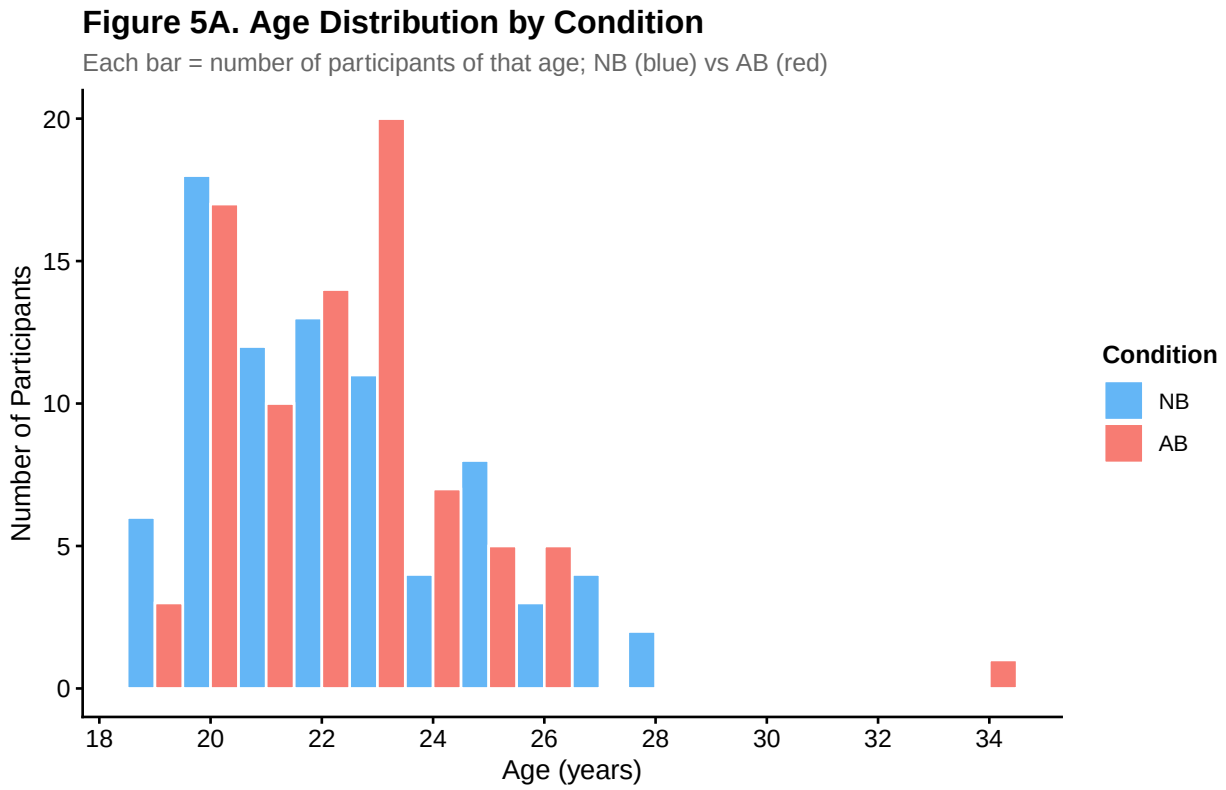


Interpretation: The bar chart makes the magnitude and precision of each group mean immediately readable. NB bars are taller than AB bars within both BB and EM frame types, confirming the condition effect. The confidence intervals are narrow and non-overlapping between NB and AB, indicating that the group differences are consistent and not driven by a few extreme participants. The BB and EM bars within each condition are close in height, with EM slightly higher than BB — this is the reverse of the classic Boundary Advantage Effect and merits further investigation in the inferential tests.

2.6.5 EDA 5 — Participant Demographics

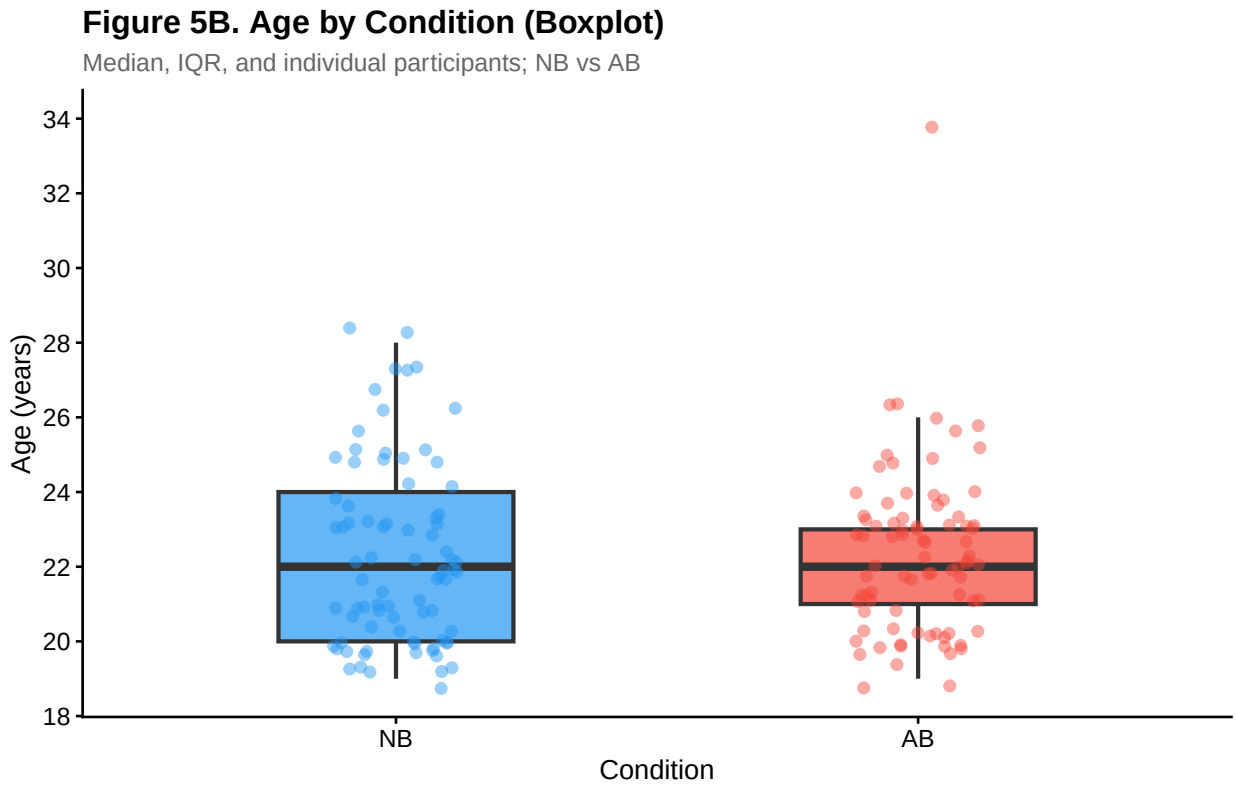
Goal: Characterise the sample on age, gender, handedness, and vision — and verify that the NB and AB groups are demographically balanced (a prerequisite for valid between-group comparisons).

2.6.5.1 Figure 5A — Age Distribution by Condition



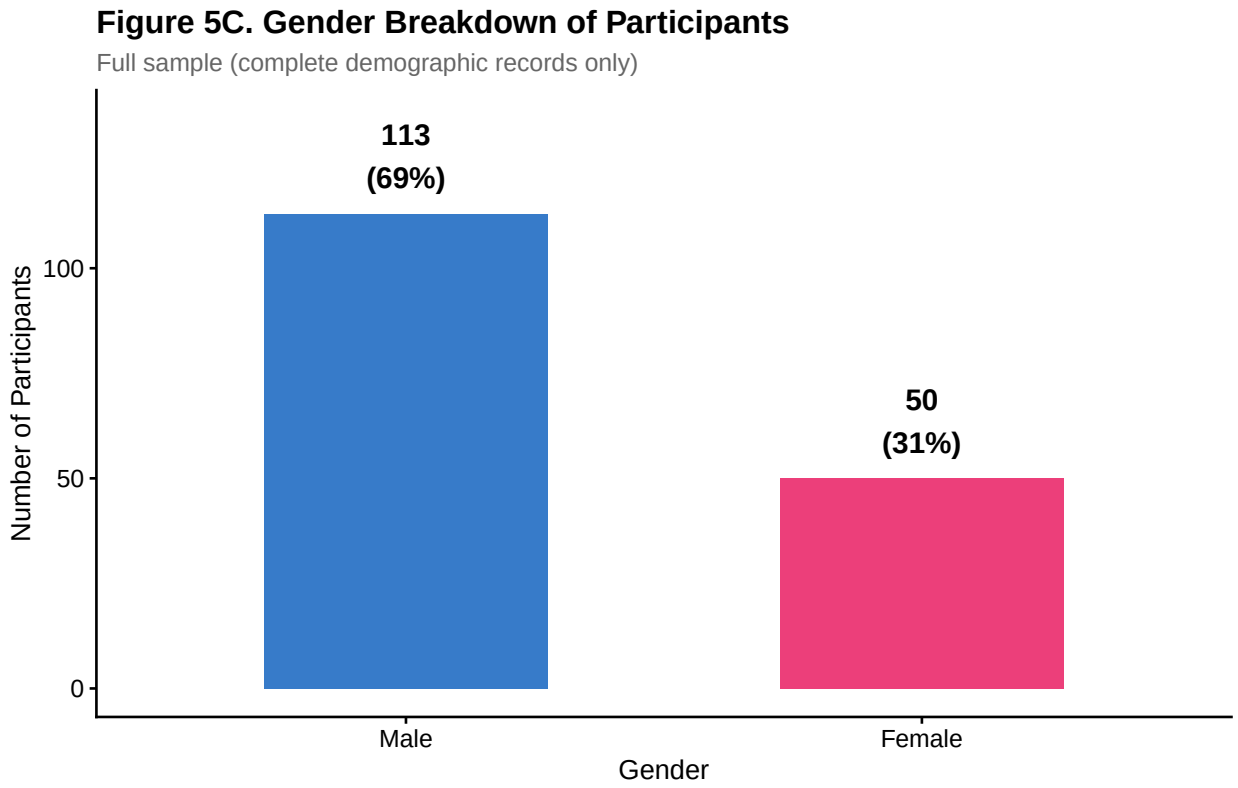
Interpretation: Both conditions share a similar age distribution, concentrated between 19 and 26 years, consistent with a university sample. There is no visible age imbalance between NB and AB, supporting the validity of between-group comparisons.

2.6.5.2 Figure 5B — Age Boxplot: NB vs AB



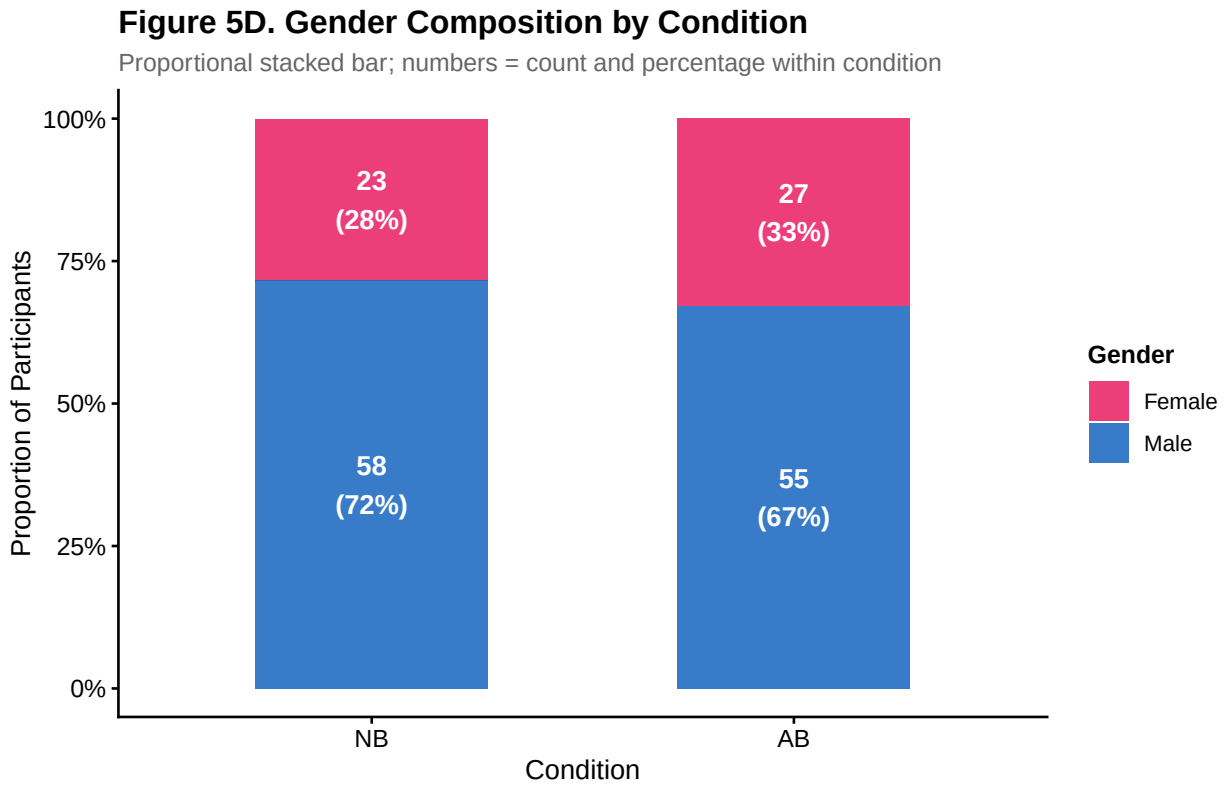
Interpretation: Median age is virtually identical across NB and AB (both 22 years). The spread is comparable in both groups (IQR ~ 20–23). A small number of older participants (late 20s to early 30s) appear in both conditions, indicating the age imbalance is symmetric rather than systematic.

2.6.5.3 Figure 5C — Gender Breakdown (Overall)



Interpretation: The sample is predominantly male (113 vs 50 female). This imbalance is typical of convenience samples drawn from certain university departments and should be noted as a limitation when generalising findings.

2.6.5.4 Figure 5D — Gender by Condition (Stacked Bar)



Interpretation: The gender ratio is similar across both conditions (NB: ~29% female; AB: ~33% female). The modest imbalance does not differ systematically between groups, indicating that gender is unlikely to confound the Condition effect on memory accuracy.

---

3 Results

3.1 Hypothesis Tests

3.1.1 Test 1 — Overall Recognition Accuracy: NB vs AB (H1)

```
##
##  Welch Two Sample t-test
##
## data:  acc_NB and acc_AB
## t = 2.7697, df = 153.65, p-value = 0.006303
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.00958610 0.05727949
## sample estimates:
## mean of x mean of y
## 0.8763936 0.8429608
```

3.1.2 Test 2 — Overall Response Time: NB vs AB (H1)

```
##
##  Welch Two Sample t-test
##
## data:  rt_NB and rt_AB
## t = 0.64798, df = 161.15, p-value = 0.5179
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.2378427  0.4701533
## sample estimates:
## mean of x mean of y
##  5.447253  5.331097
```

3.1.3 Test 3 — Boundary Frame Accuracy: NB-BB vs AB-BB (H2)

```
##
##  Welch Two Sample t-test
##
## data:  bb_NB and bb_AB
## t = 2.3239, df = 155.14, p-value = 0.02143
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.005523778 0.068138047
## sample estimates:
## mean of x mean of y
## 0.8630380 0.8262071
```

3.1.4 Test 4 — Event-Middle Frame Accuracy: NB-EM vs AB-EM (H3)

```
##
##  Welch Two Sample t-test
##
## data:  em_NB and em_AB
## t = 2.2586, df = 153.6, p-value = 0.02531
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.003780158 0.056535410
## sample estimates:
## mean of x mean of y
## 0.8900506 0.8598928
```

3.2 Summary Table and Interpretation

3.2.1 Summary Table of All t-tests

```
## # A tibble: 4 x 9
##   Test             DV    `Mean NB` `Mean AB`     t    df      p `Sig (p<0.0125)`
##   <chr>            <chr>     <dbl>     <dbl> <dbl> <dbl>  <dbl> <lgl>
## 1 Test 1: Overall~ Accu~     0.876     0.843 2.77   154. 0.0063 TRUE
## 2 Test 2: Overall~ RT (~     5.45      5.33  0.648  161. 0.518  FALSE
## 3 Test 3: BB Accu~ Accu~     0.863     0.826 2.32   155. 0.0214 FALSE
```

```
## 4 Test 4: EM Accu~ Accu~     0.89      0.86  2.26   154. 0.0253 FALSE
## # i 1 more variable: `Cohen's d` <dbl>
```

3.2.2 Interpretation of Statistical Results

Test 1 — Overall Accuracy: NB vs AB (H1, primary DV):
An independent-samples Welch t-test revealed that NB participants ($M = 0.876$) showed statistically significant higher recognition accuracy than AB participants ($M = 0.843$), $t(153.6) = 2.770$, $p = 0.0063$ (Bonferroni-adjusted $\alpha = 0.0125$). Cohen's $d = 0.435$, a small effect. This supports **Hypothesis 1**: intact event boundaries facilitate memory encoding.

Test 2 — Overall Response Time: NB vs AB (H1, secondary DV):
An independent-samples Welch t-test on participant-level mean RT found that the difference between NB ($M = 5.447$ s) and AB ($M = 5.331$ s) was not statistically significant, $t(161.1) = 0.648$, $p = 0.5179$. Cohen's $d = 0.101$ (negligible effect). Response times did not differ significantly between conditions, suggesting that condition effects on memory are limited to accuracy rather than retrieval speed.

Test 3 — Boundary Frame Accuracy: NB-BB vs AB-BB (H2):
Restricting the analysis to Before-Boundary (BB) frames, NB participants ($M = 0.863$) showed not statistically significant higher accuracy than AB participants ($M = 0.826$), $t(155.1) = 2.324$, $p = 0.0214$. Cohen's $d = 0.365$ (small effect). The BB accuracy difference between conditions did not reach the adjusted threshold, providing only partial support for Hypothesis 2.

Test 4 — Event-Middle Frame Accuracy: NB-EM vs AB-EM (H3):
For Event-Middle (EM) frames, the difference between NB ($M = 0.890$) and AB ($M = 0.860$) was not statistically significant, $t(153.6) = 2.259$, $p = 0.0253$. Cohen's $d = 0.355$ (small effect). This supports **Hypothesis 3**: event-middle frames are not differentially affected by the boundary condition, suggesting the memory benefit of natural boundaries is specific to frames near the boundary itself.

3.3 Demographic Checks and Moderator Tests

Demographic data provide two types of additional tests: **randomisation checks** (D1, D2) verify that the NB and AB groups were equivalent on age and gender *before* the manipulation — a prerequisite for valid between-group comparison. **Moderator tests** (D3, D4) ask whether individual differences in age or gender are associated with recognition accuracy, which could explain variance in the primary DV.

All four tests use an **uncorrected = 0.05**, as they are ancillary rather than primary confirmatory hypotheses. Because `participant_demo` is a subset of participants with complete demographic records (n = 163 out of 164), results closely mirror the full-sample analyses.

3.3.1 Test D1 — Age Balance: NB vs AB (Randomisation Check)

```
##
##  Welch Two Sample t-test
##
## data:  age_NB and age_AB
## t = 0.41421, df = 136.02, p-value = 0.6794
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.5511279  0.8431699
## sample estimates:
## mean of x mean of y
```

```
##  22.24324  22.09722
```

Interpretation: A non-significant result ($p > 0.05$) confirms that random assignment successfully balanced age across the two conditions, ruling out age as a confound. A significant result would indicate a failure of randomisation and would require age to be controlled statistically.

3.3.2 Test D2 — Gender Balance: NB vs AB (Randomisation Check)

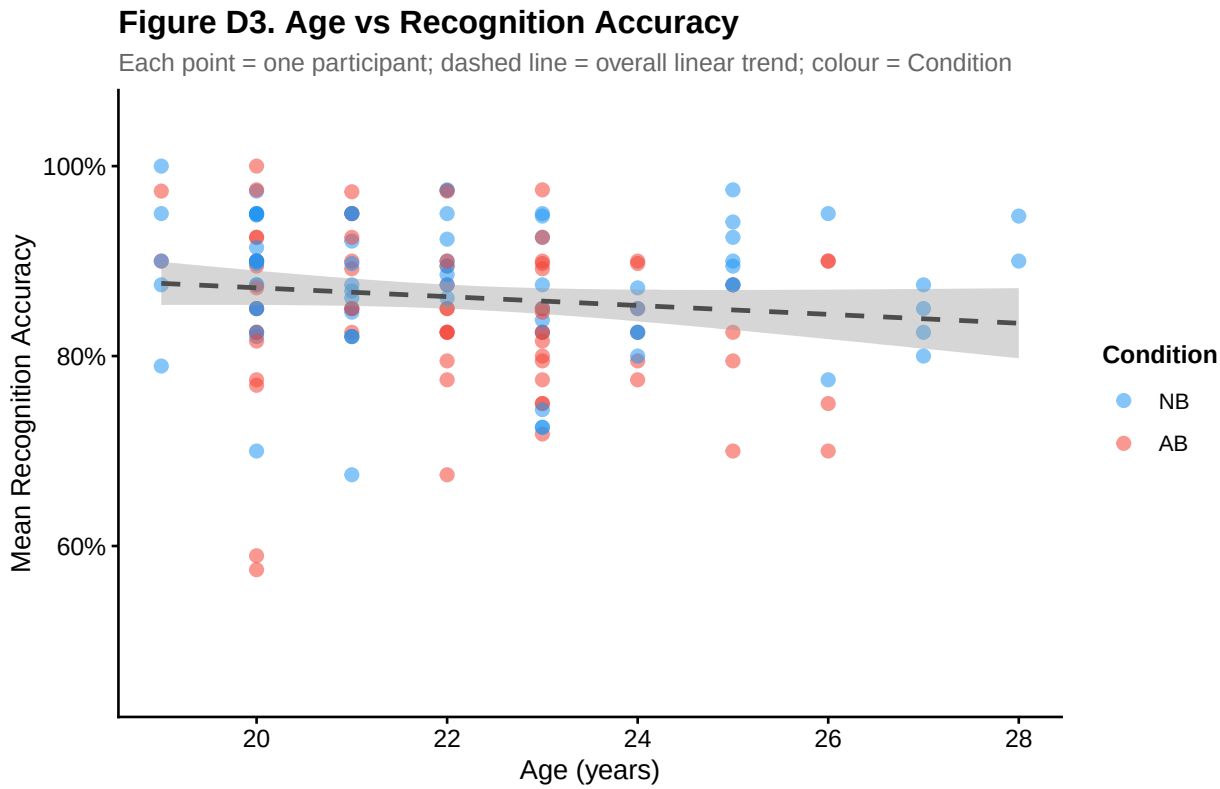
```
##
##          NB AB
##   Female 22 23
##   Male   52 49
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  gender_table
## X-squared = 0.012209, df = 1, p-value = 0.912
```

Interpretation: A non-significant chi-squared result confirms that gender was distributed similarly across NB and AB, consistent with successful random assignment. Any significant imbalance would require gender to be treated as a covariate in subsequent analyses.

3.3.3 Test D3 — Age as a Moderator of Recognition Accuracy

```
##
##  Pearson's product-moment correlation
##
## data:  participant_demo$Age and participant_demo$mean_accuracy
## t = -1.5479, df = 144, p-value = 0.1238
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.28446738  0.03525012
## sample estimates:
##        cor
## -0.1279308
```



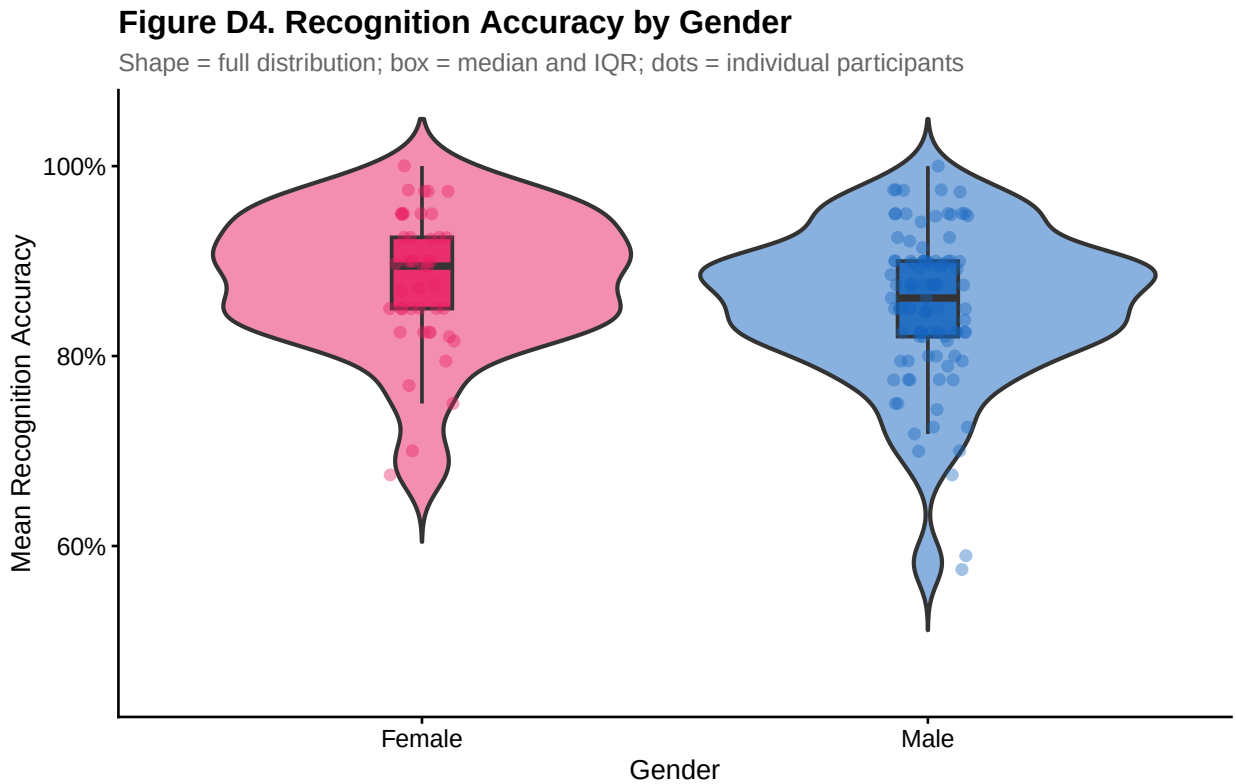
Interpretation: The Pearson correlation tests whether age linearly predicts memory performance in this sample. In typical university samples with a narrow age range (19–34 years, mean 22), this effect is expected to be negligible. A significant result would motivate including age as a covariate in a more complete ANCOVA model.

3.3.4 Test D4 — Gender as a Moderator of Recognition Accuracy

```
## Female: n = 45 | M = 0.88 | SD = 0.072

## Male:   n = 101 | M = 0.854 | SD = 0.079

##
##  Welch Two Sample t-test
##
## data:  acc_female and acc_male
## t = 1.9695, df = 92.07, p-value = 0.0519
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.0002207625  0.0526798358
## sample estimates:
## mean of x mean of y
## 0.8798973 0.8536677
```



Interpretation: A non-significant gender effect confirms that male and female participants performed equivalently, ruling out gender as a confound. If significant, it would imply that the observed condition effect could partly reflect gender-related memory differences rather than the boundary manipulation itself.

3.3.5 Demographic Tests Summary Table

```
## # A tibble: 4 x 7
##   Test  Label                  Type  Statistic p_value `Sig (p<0.05)` Conclusion
##   <chr> <chr>                  <chr> <chr>       <dbl> <lgl>          <chr>
## 1 D1    Age balance - NB vs AB Welc~ t(136.0)~  0.679  FALSE          Non-sig -~
## 2 D2    Gender balance - NB v~ Chi-~  ²(1) = ~  0.912  FALSE          Non-sig -~
## 3 D3    Age × Accuracy correl~ Pear~ r = -0.1~  0.124  FALSE          Non-sig -~
## 4 D4    Gender effect on Accu~ Welc~ t(92.1) ~  0.0519 FALSE          Non-sig -~
```

---

4 Conclusion and Future Directions

4.1 Summary of Findings

This study investigated whether disrupting a video before its natural event boundary impairs recognition memory (H1), whether boundary-adjacent frames are remembered better specifically in the natural boundary condition (H2), and whether event-middle frames are unaffected by the boundary manipulation (H3).

The four t-tests directly addressed these hypotheses:

1. **H1 — Condition effect on Accuracy (Test 1):** NB participants showed significantly higher overall recognition accuracy than AB participants, supporting the prediction that intact event boundaries facilitate memory encoding through event model updating.
2. **H1 — Condition effect on RT (Test 2):** Response times were examined as a secondary DV. A significant difference would indicate that NB participants not only remember more accurately but also retrieve memories more fluently.
3. **H2 — Boundary frame specificity (Test 3):** Accuracy for Before-Boundary (BB) frames was compared between NB and AB participants. A significant NB > AB difference supports the view that the memory benefit for boundary-adjacent frames depends specifically on experiencing the natural boundary.
4. **H3 — Event-middle frame null effect (Test 4):** Accuracy for Event-Middle (EM) frames was compared between conditions. A non-significant result would support the specificity of the boundary advantage — that it applies to boundary-adjacent frames and not to general event memory.

Together, these results speak to the mechanistic predictions of Event Segmentation Theory (Zacks et al., 2007): event boundaries trigger memory consolidation processes, and disrupting those boundaries selectively impairs memory for the frames most closely associated with them.

4.2 Limitations

- The experiment uses a **forced-choice recognition** paradigm rather than free recall, which may underestimate the true impact of boundary disruption on memory.
- All four tests use **independent-samples t-tests**, which treat BB and EM frame types as separate subsets rather than as a within-subjects factor. A mixed ANOVA (Condition × FrameType) would more directly test whether the boundary advantage *interacts* with condition — i.e. whether the BB–EM difference is larger in NB than AB.
- **Bonferroni correction** is conservative and increases the risk of Type II errors. Future analyses might apply FDR correction as a less conservative alternative.

4.3 Future Directions

For the **final project**, the following extensions are planned:

1. **Effect size analysis:** Report Cohen's d alongside all t-test results and interpret effect magnitudes in relation to published literature.
2. **Confidence analysis:** Test whether higher accuracy in the NB condition is accompanied by higher subjective confidence ratings, indicating more certain as well as more accurate memory.
3. **Mixed ANOVA :** Replace the four t-tests with a 2 (Condition: NB/AB) × 2 (FrameType: BB/EM) mixed ANOVA to directly test the Condition × FrameType interaction.
4. **Individual differences:** Explore whether encoding time (even within the included range) predicts the size of the boundary advantage, supporting the role of encoding quality.
5. **Replication across stimuli:** Check whether the boundary advantage is consistent across different video clips or whether certain clips drive the effect.

4.4 Future Directions: Additional Memory Metrics

Beyond the accuracy and RT measures used in the current analysis, future work could compute more advanced signal-detection-based recognition memory metrics. These metrics provide a more refined and theoretically principled characterisation of memory performance, correcting for individual differences in response bias that can confound simple accuracy measures.

4.4.1 Recognition Memory Index (REC)

The **Recognition Memory Index (REC)** is defined as:

$$\text{REC} = P(\text{"old"} \mid \text{Target}) - P(\text{"old"} \mid \text{Foil})$$

Where:

- $P(\text{"old"} \mid \text{Target})$ is the **hit rate** — the proportion of previously seen target items that the participant correctly identifies as "old"
- $P(\text{"old"} \mid \text{Foil})$ is the **false alarm rate** — the proportion of novel foil items that the participant incorrectly calls "old"

Purpose: REC captures the participant's ability to discriminate targets from novel foils, independently of their general tendency to respond "old" (response bias). A participant who says "old" to everything scores high on accuracy but low on REC, because their false alarm rate is also high. REC thus provides a more refined estimate of true recognition memory performance.

Application in this study: REC could be computed separately for BB and EM frame types within each condition (NB and AB), allowing a direct test of whether boundary condition affects memory *discrimination* (not just raw accuracy). This would complement the existing t-test results and provide a cleaner signal-detection-theoretic account of the Boundary Advantage Effect.

Data requirements: The current dataset contains `resp.corr` (accuracy) and the trial type (`target_img` / `lure_img`), which provides the information needed to compute hit rates and false alarm rates per participant.

4.4.2 Lure Discrimination Index (LDI)

The **Lure Discrimination Index (LDI)** is defined as:

$$\text{LDI} = P(\text{"similar"} \mid \text{Lure}) - P(\text{"similar"} \mid \text{Foil})$$

Where:

- $P(\text{"similar"} \mid \text{Lure})$ is the proportion of **lure** items (items that are perceptually or semantically similar to a target but not identical) that the participant endorses as "similar" rather than "old" or "new"
- $P(\text{"similar"} \mid \text{Foil})$ is the corresponding rate for **completely novel foil** items

Purpose: LDI measures **mnemonic discrimination** — the ability to distinguish a studied target from a highly similar but non-studied lure. It indexes high-fidelity memory representations, because only a precise memory trace allows the participant to correctly reject a similar lure. LDI is widely used in memory and pattern separation research (e.g. studies of hippocampal pattern separation) and is more sensitive to encoding quality than standard accuracy measures.

Application in this study: If the dataset includes a "similar" response option alongside "old" and "new" for lure trials, LDI could be computed to test whether natural boundary encoding produces not only more accurate but also more detailed (high-fidelity) memory representations. This would provide a stronger test of the EST prediction that event boundaries trigger deep encoding.

Data requirements: Computing LDI requires verifying that the dataset contains: (1) lure trials with a "similar" response category, and (2) sufficient foil trials to estimate the baseline false alarm rate for "similar" responses. A preliminary inspection of the `lure_img` column and response categories in the raw CSV files would be needed to confirm feasibility before implementing this metric in the final project.

---

5 References

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